Shared multi-view encoder (identical, train & inference)

5 views per image **[NEW: orientation-corrected]**
1 global (288x288) + 4 local corner crops (64% width)
the ∼9% of images flagged in `orientation_flags.json` are rotated 180° first (`fix_orientation=true`)

$5 \times 3 \times 288 \times 288$

EfficientNet-B0 RA4 backbone
– SHARED across all 5 views
`efficientnet_b0.ra4_e3600_r224_in1k`, `num_classes=0`
from ImageNet init (no warm start), 55 epochs, cosine

5×1280

Per-view features
5×1280 (1 global, 4 local)

Scene feature → low-rank cell trunk → classification/offset heads

5×1280

Fuse scene feature
$0.5 \cdot \text{global} + \text{mean}(4 \text{ local})$

1280

Dropout p=0.3
→ fused feature

1280 | 1280 | 1280

Country head
Linear $1280 \to 12$

NEW: shared cell trunk
Linear $1280 \to 224$, GELU
low-rank, shared by all 4 heads

Offset head
Linear $1280 \to 2$
(trained, but the tuned decoder sets o=0: no help)

A. Training-time losses

224

Country CE loss
weight 1.0

NEW: cell heads k=0..3
4 × Linear $224 \to 368$
368 cells/partition (was 96)
K=4 partitions (was 3)
raw **unmasked** logits

Offset regression loss
weight 0.5

NEW: haversine-smoothed cell loss (×4)
target $\propto \text{softmax}(-\text{hav}(\text{centroid}_i, \text{truth})/\tau)$
over that image's country's cells, τ=45 km
(replaces the one-hot target; PIGEON eq. 2–3)
masked to **true** country (teacher forcing)

Geographic-distribution loss
pairwise embedding cosine vs. smoothed pairwise haversine distance (scale 250 km); weight 0.2, temperature 0.1

View-consistency loss
cosine align all 5 per-view embeddings to the fused one; weight 0.1

Supervised geo-contrastive loss (from T4.1c)
geo-balanced batches (4 samples/cell); pulls together samples sharing ≥1 of the 4 partition cells; temperature 0.1, weight 0.2

From ImageNet init, 55 epochs, batch 32, backbone_lr 1e−4 / head_lr 1e−3, wd 1e−3. **EMA explicitly off** (`ema_decay=0.0`) — an earlier eval-on-EMA-weights bug invalidated two screens.
Best epoch 52/55: 137.16 km (fold-0 val median).

Embedding branch

5×1280

Embedding head (per view)
Linear $1280 \to 224$, L2-normalize

5×224

Fuse & re-normalize
$0.5 \cdot \text{emb}_{\text{global}} + \text{mean}(4 \text{ local emb})$

Fused query embedding
224-d, L2-normalized

B. Inference-time retrieval decode

Gallery (built once, offline, frozen)
every fold-*training* image's country, 4×cell_id, fused & local embeddings, unit-vector coords

Hierarchy score
$\log P(\text{country}) + \frac{1}{4}\sum_{k=0}^{3} \log P(\text{cell}_k \mid \text{country})$
(country-masked conditional log-softmax per partition)

Similarity score
$(1-\ell) \cdot \cos(q_{\text{fused}}, g_{\text{fused}}) + \ell \cdot \max_v \cos(q^v_{\text{local}}, g_{\text{local}})$
(local_weight ℓ; **best fold-0**: ℓ=0)

Combined score
$\text{hierarchy} + w \cdot \text{similarity}$
retrieval_weight w; **best swept**: w=16
(the fine-cell model wants a *much* higher w than T4.1c's w=4)

Spherical mode decode
top-m by score → pairwise great-circle distances → highest-mass coherent cluster within radius r (hard kernel) → score-weighted spherical mean
best swept: m=100, r=200 km, offset off

Predicted (lat, lng)
decimal degrees

`mode_w12_l0_r250`
(train-time grid): **137.16 km**
`mode_w16_l0_r200_hard_m100_o0`
(swept): **132.72 km**
mean 423.2 km, <200 km 0.592, <750 km 0.830